

EXP NO:2

CHARACTERISTICS STUDY OF SIDE BAND POWER AND

FREQUENCIES IN AM USING MATLAB

Objectives:

- i) To generate an Amplitude Modulated (AM) signal in MATLAB using specified carrier and modulating signal parameters.
- ii) To determine and verify the upper sideband (USB) and lower sideband (LSB) frequencies of the AM signal using frequency spectrum analysis in MATLAB.
- iii) To calculate the carrier power, sideband power, and total transmitted power of the AM signal for a given modulation index using MATLAB.
- iv) To analyze the effect of varying the modulation index on sideband power distribution and overall transmission efficiency through MATLAB simulation.

CODE

```
clc;
clear;
close all;

% Parameters
Am = 1;
fm = 500;
Ac = 5;
fc = 5000;
fs = 20*fc;

% Time vector
t = 0:1/fs:4/fm;

% Message signal
m = Am*sin(2*pi*fm*t);

% Carrier signal
c = Ac*sin(2*pi*fc*t);

% Modulation index
mu = Am/Ac;

% AM signal
am = Ac*(1 + mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);

% Demodulation using rectification and moving average
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rectified = abs(am);
window = max(1, round(fs/fc));
dem = movmean(rectified, window);
dem = dem - mean(dem);

% Frequency spectrum
N = length(am);
f = (-floor(N/2):ceil(N/2)-1)*(fs/N);
AM_FFT = abs(fftshift(fft(am)))/N;

% Sideband frequencies
USB = fc + fm;
LSB = fc - fm;

% Power calculations (1-ohm load)
Pc = Ac^2/2;
Psb = (mu^2*Pc)/4;

% Plot results
figure('Name','AM Modulation','NumberTitle','off');

subplot(3,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,3);
plot(t,am);
title('AM Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,4);
plot(t,dem);

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title('Demodulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,2,5);
plot(f,AM_FFT);
xlim([fc-3*fm fc+3*fm]);
title('AM Frequency Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;

subplot(3,2,6);
bar([Pc Psb Psb]);
set(gca,'XTick',1:3);
set(gca,'XTickLabel',{'Carrier','USB','LSB'});
title('Power Distribution');
ylabel('Power (W)');
grid on;

% Display results
fprintf('Modulation index = %.2f\n',mu);
fprintf('USB frequency = %d Hz\n',USB);
fprintf('LSB frequency = %d Hz\n',LSB);
fprintf('Carrier power = %.2f W\n',Pc);
fprintf('Power per sideband = %.3f W\n',Psb);
fprintf('Total transmitted power = %.3f W\n',Pc+2*Psb);

```

OUTPUT

